Julio M. Morales

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Research Profile

Fifth-year Ph.D. candidate in Astronomy specializing in atmospheric gravity waves and their coupling to and conversion into magnetohydrodynamic (MHD) wave modes in the lower solar atmosphere. My research combines theoretical modeling, high-resolution multiwavelength data, and three-dimensional radiative MHD simulations to investigate how magnetic-field geometry, radiative damping, wave reflection, and mode conversion influence observable wave signatures. This work characterizes wave propagation and energy transport in the solar atmosphere and explores how these observable signatures can be used to infer atmospheric properties, wave dynamics, and contribute to the chromospheric energy budget.

Education

New Mexico State University — Expected 2027
Ph.D. in Astronomy; Advisor: Prof. Jason Jackiewicz — Las Cruces, NM

New Mexico State University — May 2025
M.S. in Astronomy; College of Arts and Sciences — Las Cruces, NM

University of Massachusetts Amherst — May 2022
B.S. in Physics and Astronomy; College of Natural Sciences — Amherst, MA

Research Experience

Ph.D. Candidate and Graduate Research Assistant — Aug 2022 - Present
New Mexico State University — Las Cruces, NM
Advisor: Prof. Jason Jackiewicz

- Investigate atmospheric gravity waves and their interactions with magnetic fields in the lower solar atmosphere.
- Combine theoretical modeling, solar observations, and numerical simulations to study wave propagation and energy transport.
- Analyze solar wave signatures to explore their use as diagnostics of atmospheric conditions.
- Develop scientific software for data analysis, numerical modeling, and visualization.
- First author of an in-progress manuscript on theoretical and observational diagnostics of atmospheric gravity waves in the lower solar atmosphere.
- Co-authored peer-reviewed research on solar waves and magnetic fields.

Post-Baccalaureate Researcher — Jul 2022 - Sep 2022
Amherst College — Amherst, MA
Advisor: Prof. Katherine B. Follette

- Continued analysis of second-to-minute timescale Hα variability in transitional disk host-stars.
- Advanced thesis-derived accretion variability research for manuscript development within the Giant Accreting Protoplanet Survey.
- Presented this research at the international Cool Stars Conference in Toulouse, France (July 2022).

Senior Thesis Researcher Sep 2021 - May 2022
University of Massachusetts Amherst Amherst, MA
Advisor: Prof. Katherine B. Follette
Thesis: *Accretion Variability in Transitional Disk Host-Stars: Second-to-Minute Timescale Hα Variability in the Giant Accreting Protoplanet Survey*

- Investigated short-timescale Hα variability as a tracer of accretion variability in transitional disk host-stars.
- Presented thesis results at departmental, regional, and national astronomy meetings.

Five College Astronomy Department Summer Researcher Jul 2021 - Sep 2021
Amherst College Amherst, MA
Advisors: Prof. Katherine B. Follette and Dr. Connor Robinson

- Contributed to Giant Accreting Protoplanet Survey research on accreting protoplanets and disk-host star variability.
- Presented research at the Amherst College Summer Research Poster Session and Five College Astronomy Department Research Presentations.

Publications

Morales, J. M., Jackiewicz, J., Vesa, O., & Vigeesh, G. 2027 (in progress), *Connecting Theoretical and Observational Phase-Difference Diagnostics of Atmospheric Gravity Waves in the Lower Solar Atmosphere.*

Vesa, O., **Morales, J. M.**, Jackiewicz, J., Vigeesh, G., & Reardon, K. 2025, *ApJ*, 992, 201, Atmospheric Gravity Waves Modulated by the Magnetic Field Configuration.

Follette, K. B., Close, L. M., Males, J. R., Ward-Duong, K., Balmer, W. O., **Morales, J. M.**, et al. 2022, arXiv:2211.02109, The Giant Accreting Protoplanet Survey (GAPlanetS).

Balmer, W. O., Follette, K. B., Close, L. M., Males, J. R., De Rosa, R. J., **Morales, J. M.**, et al. 2022, arXiv:2206.00687, Improved Orbital Constraints and Hα Photometric Monitoring of HD 142527 B.

Morales, J. M., Balmer, W. O., & Follette, K. B. *Hα Variability in the Giant Accreting Protoplanet Survey.* Manuscript in preparation.

Conferences & Presentations

Waves and Instabilities in the Solar Atmosphere, Monte Porzio Catone, Italy. Presented.	Oct 2024
SACNAS Diversity in STEM Conference, Phoenix, AZ. Attended.	Oct 2023
SACNAS Diversity in STEM Conference, San Juan, PR. Presented.	Oct 2022
2022 Northeast Star and Planet Formation Meeting, Middletown, CT. Attended.	Jul 2022
Cool Stars Conference, Toulouse, France. Presented.	Jul 2022
American Astronomical Society Meeting 240, Pasadena, CA. Presented.	Jun 2022
Amherst College Physics and Astronomy Senior Thesis Talk, Amherst, MA. Presented.	May 2022
Five College Astronomy Department Thesis Picnic Talk, Amherst, MA. Presented.	May 2022
SACNAS Diversity in STEM Conference, Virtual. Attended.	Nov 2021
Amherst College Summer Research Poster Session, Amherst, MA. Presented.	Sep 2021
Five College Astronomy Department Research Presentation, Amherst, MA. Presented.	Aug 2021

Teaching & Mentorship

Private Tutor Fall 2024 - Present
Independent

- Provide one-on-one instruction in Algebra I–II, Precalculus, Trigonometry, Calculus I–II, Introductory Statistics, Physics I, Introductory Astronomy, Observational Astronomy, Modern Astrophysics, and Numerical Methods.

Teaching Assistant, Summer Science Program in Astrophysics Summer 2024
University of North Carolina at Chapel Hill Chapel Hill, NC

- Tutored advanced international high-school students in physics, calculus, and Python.
- Delivered instruction in observational astronomy and trained students on research-grade telescopes.
- Supervised asteroid-tracking data acquisition.
- Modeled professional academic communication, collaboration, and networking practices.

FRIENDS Mentorship Program Jan 2023 - May 2025
Astronomy Graduate Student Organization, NMSU Las Cruces, NM
Liaison to Undergraduate Affairs

- Redesigned FRIENDS into a structured research-preparatory mentorship pipeline for undergraduate astronomy students.
- Mentored undergraduate Daniela Reyes in research readiness, expectations, accountability, and academic communication.
- Developed mentor-mentee agreements and individualized development plans to support goal-setting and follow-through.

Introduction to Astronomy Aug 2022 - Present
New Mexico State University Las Cruces, NM
Lab Instructor

- Lead undergraduate astronomy laboratory instruction and support students in applying quantitative and observational concepts.

Astronomy Help Desk 2021 - 2022
University of Massachusetts Amherst Amherst, MA
Tutor

- Tutored undergraduates in astronomy problem-solving, quantitative reasoning, and course preparation.

TRIO Upward Bound 2019 - 2020
University of Massachusetts Boston Boston and Paxton, MA
Instructor, Tutor, and Teaching Assistant

- Supported college-preparatory STEM instruction for first-generation and low-income high-school students.

Service & Leadership

Inclusive Astronomy Coordinator Jan 2023 - May 2025
Astronomy Graduate Student Organization, NMSU Las Cruces, NM

- Led restructuring of Inclusive Astronomy into an action-oriented equity initiative within the astronomy department.
- Designed a department-wide Individual Development Plan framework for graduate-student career planning and mentorship accountability.

AstroBetter Contributor Oct 2023 - Feb 2024
Scholarly Communication

- Authored a five-part series on scientific workflows: Google Calendar, Zotero, Git/GitHub, Visual Studio Code, and Notion.

Teaching Academy Aug 2022 - Present
New Mexico State University Las Cruces, NM
Sustaining Member

Letters to a Pre-Scientist Aug 2022 - Present
Scientist Pen-Pal

Hidden Figures Workshop 2020
TRIO Upward Bound, UMass Boston Boston, MA
Founder and Facilitator

Outreach

- Completed 12+ hours of AGSO STEM outreach at New Mexico State University.
- Observing Assistant, Star Party, Las Cruces, NM, Oct 2022.
- Panelist, Research Experience for Undergraduates Q&A Panel, Amherst, MA, Sep 2021.
- Outreach Coordinator, *Explore the Solar System: Amazing Facts & History of the Solar System*, Northampton, MA, Jul 2021.
- Observing Night Assistant, Modern Astronomy Summer Pre-Course, Amherst, MA, Jul 2021.

Technical & Observational Skills

Programming	Python; Fortran; MATLAB; JavaScript; Bash; Zsh; Git/GitHub; LaTeX; HTML.
Scientific Computing	Solar and stellar data analysis; scientific visualization; reproducible research workflows; documentation and version control.
AI Workflows	Applied LLMs for research workflow automation, scientific data visualization, documentation, and version-controlled development.
Software Development	Developer of AstroStack, a research workflow platform for scientific project organization and automation.
Certified Observer	Tortugas Mountain Observatory; Apache Point Observatory.
Observing Experience	Southern Astrophysical Research (SOAR) Telescope, Cerro Pachón, Chile; Amherst College Observatory, Amherst, MA.

Awards & Honors

New Mexico State University Diversity Award Scholarship. May 2023
AAS Chambliss Astronomy Achievement Award Competition Medalist. 2022
Webber Voyager Graduate Fellowship, $7,500/yr. 2022
SACNAS Travel Scholarship. 2022
UMass Amherst Dean's List Honors. 2021 - 2022
John and Abigail Adams Scholarship, $1,400/yr. 2017

Languages

English, native fluency | Spanish, native proficiency | Arabic, beginner